Pedro Morales

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Data Scientist and Mathematician with proficiency in data processing, feature engineering, machine learning, and statistical modeling. Seeking a full-time position as a Data Scientist starting July 2027.

Education

Purdue University — West Lafayette, IN
Ph.D. in Mathematics — *Expected May 2027*

University of Maryland, College Park — College Park, MD
B.S. in Mathematics — *May 2021*

Technical Skills

Programming: Python, SQL (BigQuery, MySQL), R, MATLAB.

Data Science: pandas, NumPy, Polars, scikit-learn, Matplotlib, Seaborn, dplyr, ggplot2, Parquet, DuckDB, Google Colab.

Quantitative Methods: Statistical Inference, Machine Learning, Exploratory Data Analysis, Predictive Modeling, Feature Engineering, Data Visualization, Time Series Analysis, Hypothesis Testing, Regression.

Cloud & Data: Google Cloud Platform (BigQuery, Cloud Storage, Dataplex, Earth Engine), Terraform.

Developer Tools: Git, GitHub, Latex, Jupyter.

Data Science Experience

Climate Sensor and Satellite Data Analysis | **The Data Mine – Purdue University** — August 2025 - May 2026
Graduate Data Science Researcher | *Python, Google Cloud Platform, Terraform* — *West Lafayette, IN*

- Built an automated cloud ingestion pipeline (Python, SQL, BigQuery, Earth Engine) for mangrove satellite data, cutting runtime to under 30 seconds and eliminating duplicate records.
- Analyzed satellite and climate sensor data with the World Wildlife Fund to inform mangrove conservation strategy; delivering an ordered data infrastructure and presenting results to WWF directors.
- Cataloged 13 multi-source datasets in Google Cloud (BigQuery, Cloud Storage, Dataplex) into raw, cleaned, and production-ready stages, reducing analyst search time and enforcing consistent data quality.

AC Transit On Time Performance Analysis — September 2026 - October 2026
Independent Project | *Python, polars, DuckDB, SQL* — *Lafayette, IN*

- Processed a 1,440GB open GTFS-RT archive (100M vehicle pings) with DuckDB, SQL, and polars into a stop arrival and departure dataset for statistical analysis.
- Designed a bronze/silver/gold Parquet data model (DuckDB, polars, SQL) with deduplication, time-zone normalization, and handling of missing stop data.
- Analyzed on-time performance (polars, NumPy, Seaborn, Matplotlib), finding the percentage of buses that ran on time (within 5 minutes of schedule) showing the biggest reason of delay.

YouTube Trending Channel Analysis — June 2026 - August 2026
Independent Project | *Python, pandas, NumPy, scikit-learn* — *Lafayette, IN*

- Cleaned and aggregated 105k+ video records from a Kaggle dataset, sampled from a 20GB dataset into a 19k+ channel dataset using Python (pandas, DuckDB), preparing data for statistical analysis and machine learning modeling.
- Trained and evaluated scikit-learn machine learning models on imbalanced data (17.3% positive class), raising precision from 0.367 to 0.581 (58% relative gain) using cross-validation.
- Engineered features and analyzed upload frequency (NumPy, Matplotlib, Seaborn), finding that moving from weekly to daily uploads was associated with only a 1.8 percentage point gain in trending likelihood.

Quantitative Research Experience

Purdue University — August 2022 - Present
Doctoral Researcher — *West Lafayette, IN*

- Formulated and solved open-ended quantitative problems, breaking them into tractable pieces and validating results against theoretical benchmarks.
- Built mathematical models of wave propagation and derived asymptotic approximations to predict long-time system behavior, translating abstract theory into precise, testable results.
- Generalized findings from specific cases to broader frameworks, identifying the core structure of a problem so solutions apply across a wider class of settings.
- Collaborated with researchers at 3 institutions and communicated results through 1 paper, 1 manuscript in preparation, and 2 talks to both specialist and non-specialist audiences.

Technical Communication & Writing Experience

Presentations

- Presented satellite and climate data analysis to World Wildlife Fund directors and corporate partners, translating technical results into conservation strategy at the *Corporate Partners Symposium* at Purdue University. May 2026
- Presented research on unconditional wave decay in dimension two to a specialist audience at the *SIAM Advancements in Inverse Problems*. October 2026

Writing

- Delivered a literature review on network optimization for a WWF ManglarIA research paper in progress, synthesizing challenges and open questions identified by other researchers.
- Wrote a research proposal outlining doctoral progress, defining future objectives, and making the case for the work's significance.

Leadership & Mentorship Experience

Teaching Assistant August 2022 - Present

- Led weekly recitations for 50–80 undergraduates and two flipped-classroom sections (45 students), breaking down complex concepts; earned course-evaluation ratings of 4.6/5 for clear communication and 4.7/5 for answering questions.
- Held office hours and graded assignments with detailed feedback to support individual student learning.

Seminar Organization August 2023 - May 2024

- Co-organized the Graduate Student Analysis Seminar at Purdue University, coordinating speakers and weekly presentations to promote research discussions and collaboration among graduate students.

Graduate Student Mentor August 2023 - May 2025

- Mentored three prospective mathematics graduate students through the Association for Women in Mathematics (AWM) mentorship program at Purdue University, providing guidance on graduate study and academic development.

Awards

Purdue Alliance Scholar Fellowship August 2022 - May 2024

- Competitive fellowship awarded to top incoming graduate students at Purdue University, featuring two years of full financial support.

Barbara A. Kunze Award May 2026

- Awarded by the Mathematics Department at Purdue University for an outstanding research proposal and doctoral progress.